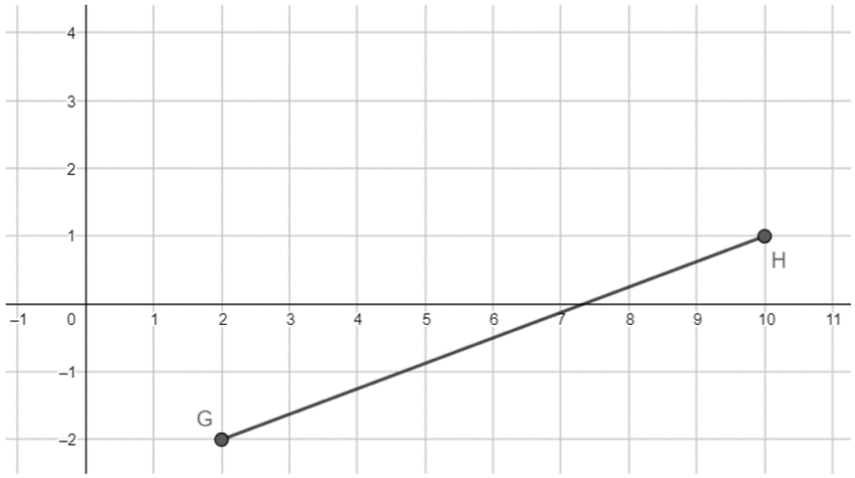
Geometry
Unit 13
Lesson 5 Homework

Name___Answer Key_______
Date_____________________
Period____________________

Use the following information for questions 1-2. Point G has coordinates $(2,-2)$ and point H has coordinates $(10,1)$.



1. Complete the table to partition $\overline{GH}$ in a $3:1$ ratio.

	x-value	y-value
$G(2,-2)$	$\frac{1}{4}(2)=\frac{1}{2}$	$\frac{1}{4}(-2)=-\frac{1}{2}$
$H(10,1)$	$\frac{3}{4}(10)=7\frac{1}{2}$	$\frac{3}{4}(1)=\frac{3}{4}$
Total	8	$\frac{1}{4}$

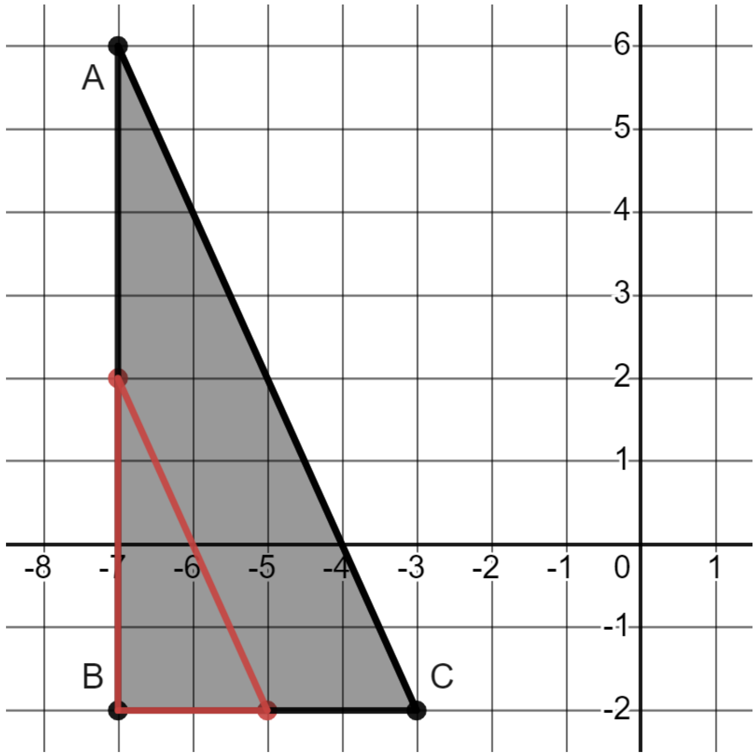
2. What is the midpoint of $\overline{GH}$?
$\left(\frac{2+10}{2},\frac{-2+1}{2}\right)=\left(6,-\frac{1}{2}\right)$

Use the following information for questions 3-4. Line segment BT has endpoints at $B(-6,-4)$ and $T(4,6)$.

3. Write an expression for the point that partitions directed line segment BT in a $2:3$ ratio.
$\frac{3}{5}(\text{point } B)+\frac{2}{5}(\text{point } T)$

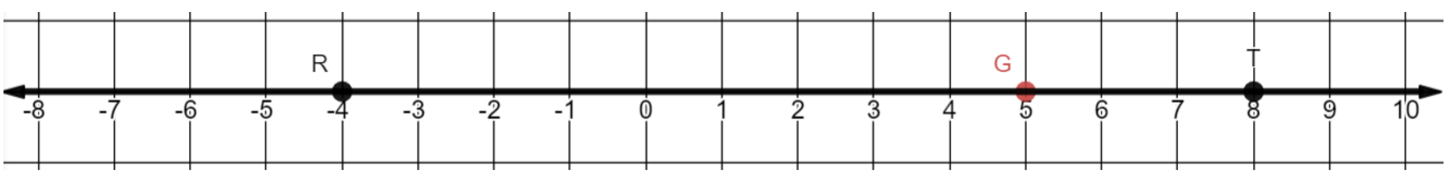
4. Use the expression to find the point that partitions directed line segment BT in a $2:3$ ratio.
$\left(\frac{3}{5}(-6)+\frac{2}{5}(4),\frac{3}{5}(-4)+\frac{2}{5}(6)\right)=(-2,0)$

Use the following information for question 5. Triangle ABC has vertices at $A(-7,6)$, $B(-7,-2)$, and $C(-3,-2)$.



5. Use segment partitioning to dilate triangle ABC using center B and scale factor $\frac{1}{2}$.

Use the following information for questions 6-7. A number line is shown. Each grid space is 1 unit.



6. Determine the point on the number line that satisfies the expression shown. Label the point G.

$$\frac{1}{4}(\text{point } R) + \frac{3}{4}(\text{point } T)$$

$$\frac{1}{4}(-4) + \frac{3}{4}(8) = 5$$

7. What is the ratio of $RG : GT$?

$5 - (-4) : 8 - 5$

$9 : 3$